THE ASSAM
ROYAL GLOBAL UNIVERSITY
GUWAHATI

ROYAL SCHOOL OF PHYSICAL EDUCATION & SPORTS

(RSPES)

DEPARTMENT OF PHYSICAL EDUCATION

COURSE STRUCTURE & SYLLABUS

(BASED ON NATIONAL EDUCATION POLICY 2020)

(4 YEARS SINGLE MAJOR)

W.E.F

AY - 2026 – 27

STRUCTURE OF THE SYLLABUS FOR 4-YEAR UG PROGRAMME (FYUGP)

ROYAL SCHOOL OF PHYSICAL EDUCATION AND SPORTS

PROGRAMME: BACHELOR OF PHYSICAL EDUCATION AND SPORTS

1st SEMESTER							
Component	**COURSE CODE**	**NAME OF PAPER**	**Level**	**L**	**T**	**P**	**CREDIT**
Major	BPE292M101	History & Philosophy of Physical Education	100	3	1		4
Major	BPE292M102	Anatomy And Physiology	100	3	1		4
Major	BPE292M111	**Sports Activity** (Volleyball, Football, Cricket)	100			8	4
Skill Enhancement Course	BPE292S111	Formal Activities, Calisthenics, Marching	100			6	3
Ability Enhancement Course	CEN982A101	Communicative English: Listening Skill	100	2			2
Value Added Course	BPE292V101	Health & Wellness	100	2			2
IKS	BPE292K101	Indian Knowledge system & Traditional Games	100	2			2
		TOTAL CREDIT FOR 1st SEMESTER					21

2nd SEMESTER							
Component	**COURSE CODE**	**NAME OF PAPER**	**Level**	**L**	**T**	**P**	**CREDIT**
Major	BPE292M201	Exercise Physiology	100	3	1		4
Major	BPE292M202	Health Education & Environmental Studies	100	4			4
Major	BPE292M211	**Sports Activity** (Athletics (Running Events), Table Tennis, Pickleball)	100			8	4
Skill Enhancement Course	BPE292S211	Yoga Techniques and Training	100			4	2
Ability Enhancement Course	CEN982A201	Communicative English	100	2			2
Value Added Course	BPE292V201	Holistic Yoga Education	100	2			2

IKS	BPE292K201	Introduction To the Indian Knowledge System	100	2			2
Swayam	SWAYAM CODE 1	Swayam 1	100				3/4
		TOTAL CREDIT FOR 2nd SEMESTER					**20+ 3/4**

3rd SEMESTER							
Component	**COURSE CODE**	**NAME OF PAPER**	**LEVEL**	**L**	**T**	**P**	**CREDIT**
Major	BPE292M301	Foundation Of Sports Training	200	4			4
Major	BPE292M302	Sports Sociology	200	4			4
Major	BPE292M311	**Sports Activity** Athletics (Jumping & Throwing Events), Basketball	200			8	4
Skill Enhancement Course	BPE292S311	Teaching Practice (General Lesson Plan)	200			4	2
Ability Enhancement Course	CEN982A301	Communicative English	200	2			2
Value Added Course	BPE292V301	Olympic Movements	200	2			2
IDC	BPE292I301	Sports Nutrition	200	2			2
Swayam	SWAYAM CODE 2	Swayam 2	200				3/4
		TOTAL CREDIT FOR 3rd SEMESTER					20+ 3/4

4th SEMESTER							
Component	**COURSE CODE**	**NAME OF PAPER**	**LEVEL**	**L**	**T**	**P**	**CREDITS**
Major	BPE292M401	Kinesiology & Biomechanics	200	4			4
Major	BPE292M402	Organisation & Administration in Physical Education	200	3	1		4
Major	BPE292M411	**Sports Activity** {Swimming, Badminton, Lawn Tennis, Squash}	200			8	4

Major	BPE292M403	ICT In Physical Education	200	3		2	4
Skill Enhancement Course	BPE292S411	Teaching Practice (Specific Lesson Plan)	200			4	2
Ability Enhancement Course	CEN982A401	Communicative English	200	2			2
Swayam	SWAYAM CODE 3	SWAYAM 3	200				3/4
		TOTAL CREDIT FOR 4TH SEMESTER					20+ 3/4

INTERNSHIP

- **Internship shall be adhered to by all students, consisting of 4 credits, which is 120 hrs.**
- **Usually of 1 month during the summer vacation between 4th & 5th Semester.**
- **The 4 credits of internship are to be evaluated as per the scheme of evaluation in the 5th Semester.**

5th SEMESTER							
Component	**COURSE CODE**	**NAME OF PAPER**	**LEVEL**	**L**	**T**	**P**	**CREDITS**
Major	BPE292M501	Sports Psychology	300	4			4
Major	BPE292M502	Test, Measurement and Evaluation	300	3	1		4
Major	BPE292M511	**Sports Activity** {Kabaddi, Kho Kho, Handball, Hockey}	300			8	4
Major	BPE292M503	Adapted Physical Education	300	4			4
Major	BPE292M504	Officiating & Coaching Basics	300	4			4
Internship	BPE292M521	Internship	300				4
Swayam	SWAYAM CODE 4	Swayam 4	300				3/4
		TOTAL CREDIT FOR 5TH SEMESTER					24+3/4

6TH SEMESTER							
Component	**COURSE CODE**	**NAME OF PAPER**	**LEVEL**	**L**	**T**	**P**	**CREDITS**
Major	BPE292M601	Methods In Physical Education	300	4			4

Major	BPE292M602	Physiotherapy And Rehabilitation in Sports	300	3	1		4
Major	BPE292M611	**Sports Activity** {Judo, Boxing, Weightlifting, Gymnastics}	300			8	4
Major	BPE292M612	Classroom Teaching & Leadership Camp	300			8	4
Major	BPE292M603	Professional Preparation in Physical Education	300	4			4
		TOTAL CREDIT FOR 6TH SEMESTER					20

7TH SEMESTER							
Component	**COURSE CODE**	**NAME OF PAPER**	**LEVEL**	**L**	**T**	**P**	**CREDITS**
Major	BPE292M701	Educational Technology	400	4			4
Major	BPE292M702	Management In Physical Education and Sports	400	4			4
Major	BPE292M711	Specialisation (Practical)	400			8	4
Major	BPE292M703	Sports Journalism	400	4			4
Major	BPE292M704	Personality Development and Communication Skill, Seminar	400	3		2	4
		TOTAL CREDIT FOR 7TH SEMESTER					20

7TH SEMESTER SPECIALISATION

- **Specialisation : Football, Cricket, Badminton, Table Tennis, Volleyball, Yoga (any one sport)**

8TH SEMESTER							
Component	**COURSE CODE**	**NAME OF PAPER**	**LEVEL**	**L**	**T**	**P**	**CREDITS**

Major	BPE292M821	Research Project/Dissertation	400				12
Major	BPE292M801	Research Methodology in Physical Education	400	4			4
Major	BPE292M802	Applied Statistics in Physical Education & Sports	400	4			4
		TOTAL CREDIT FOR 8TH SEMESTER					20

- FOR THE STUDENTS WHO ARE NOT ELIGIBLE FOR THE RESEARCH PROJECT

COMPONENT	**COURSE CODE**	**NAME OF PAPER**	LEVEL	L	T	**p**	**CREDITS**
MAJOR	BPE292M803	Cognitive Processes in Sports Performance	400	4			4
MAJOR	BPE292M804	Injury Mechanics and Prevention in Sports	400	4			4
MAJOR	BPE292M805	Ageing, Exercise and Health Physiology	400	4			4

DETAILED SYLLABUS OF 1ST SEMESTER

MAJOR SUBJECT: HISTORY & PHILOSOPHY OF PHYSICAL EDUCATION
SUBJECT CODE: BPE292M101 **L-T-P = 3-1-0,**
CREDIT UNITS: 04 **SCHEME OF EVALUATION: THEORY (T)**

Course Objective: - The course aims to provide students with a comprehensive understanding of the historical evolution, philosophical foundations, and conceptual framework of Physical Education. It focuses on developing knowledge about allied sciences, global and Indian perspectives, and the principles of physical fitness, enabling students to critically analyse the role and significance of Physical Education in modern society.

CO	Course Outcome	Bloom's Taxonomy Level
CO 1	**Recall** the meaning, scope, aims, objectives, and fundamental concepts of Physical Education	**BT 1**

CO 2	**Explain** the contribution of allied sciences and basic concepts related to Physical Education	**BT 2**
CO 3	**Apply** the principles of physical fitness, exercise, and growth & development in practical situations	**BT 3**
CO 4	**Analyze** the historical development and philosophical foundations of Physical Education at global and national levels	**BT 4**

Modules	Course Content	Periods
I	**INTRODUCTION** 1.1 Meaning and definition of Education and Physical Education 1.2 Scope, Misconception and Modern Trends in Physical Education 1.3 Aims and Objectives of Education and Physical Education 1.4 Relationship of Physical Education with General Education 1.5 Physical Education as a Profession. 1.6 Growth and Development: Meaning, Importance and Stages of Development.	**15**
II	**PHYSICAL EDUCATION WITH ALLIED CONCEPTS** 2.1 Physical Education as an Art and Science 2.2 Contribution of Allied Science to Health Education – Anatomy, Physiology, Kinesiology, Sports Medicine, Sports Biomechanics, Sports Sociology, Sports Psychology, Anthropometry and Sports Nutrition. 2.3 Meaning of Important Terms in Physical Education – Athletics, Games and Sports, Gymnastics, Formal Activities, Aquatics, Yoga, Rhythmic, Recreation, Camping, Leadership, Hiking, Minor Game, Adapted Physical Education, Physical Culture, Plyometrics, Drill, Marching, Indigenous Activities etc. 2.4 Different Ages – a) Chronological b) Physiological c) Anatomical d) Mental	**15**
III	**ORIGIN AND DEVELOPMENT OF PHYSICAL EDUCATION** 3.1 History of Physical Education in – Greece, Rome and Egypt. 3.2 Development of Physical Education during the 20th Century with special reference to – Germany, Sweden, Denmark, USSR, USA, Great Britain and Japan. 3.3 Philosophical Foundations: Idealism, Pragmatism, Naturalism, Realism, Humanism and Existentialism. 3.4 Definition of Physical Fitness, General benefits of exercise, Basic principles of exercise and physical fitness	**15**
IV	**DEVELOPMENT OF PHYSICAL EDUCATION IN INDIA AND GLOBAL EVENTS** 4.1 History and development of Physical Education in India from Ancient Time to Modern Age – Vedic Period, Epic Age, Historical Age, Nalandian Period, Mughal Period, British Period and Post-Independence Period. 4.2 Contribution of Y.M.C.A, Akhadas and Vyamsalas in development of Physical Education. 4.3 Nature and system of Physical Education in Ancient, Medieval and Modern India. 4.4 Olympic Movement – Ancient and Modern Olympics, International Olympic Committee (IOC) and its functions, Commonwealth Games, Asian Games, Afro-Asian Games,	**15**

	4.5 Sports Bodies in India – National Sports federations, Sports Schemes and Sports Awards in India, All India Council of Sports, Association of Indian Universities, National Institute of Sports, Sports Authority of India (SAI), Lakshmibai National Institute of Physical Education.	
	Total	**60**

CREDIT DISTRIBUTION		
LECTURE/TUTORIAL	**PRACTICALS**	**EXPERIENTIAL LEARNING**
60	00	30

References:

1. Kamlesh, M.L., *"Principles and History of Physical Education".* Tandon Publications, Ludhiana.

2. Singh, A., *"Modern Textbook of Physical Education, Health and Sports".* Kalyani Publishers. New Delhi.

3. Butler, George D. *Introduction to Community Recreation (5th Ed.)* New York: McGraw Hill.

4. Fitzgerald, Gerald B. "Leadership in Recreation". New York, A.S. Barnes and Co. 1941.

5. Meyar, Herold D. & Bright, Bill, Charles K. "*Community Recreation*", Boston D.C. Health Co. 1948.

MAJOR SUBJECT: ANATOMY & PHYSIOLOGY
SUBJECT CODE: BPE292M102 **L-T-P = 3-1-0,**
CREDIT UNITS: 04 **SCHEME OF EVALUATION: THEORY (T)**

Course Objective: - The course aims to provide a comprehensive understanding of the structure and functions of the human body and their application in Physical Education and Sports. It focuses on developing foundational knowledge of anatomical and physiological systems, including skeletal, muscular, and functional body systems, enabling students to understand body mechanisms, improve physical performance, and promote overall health and well-being.

CO	Course Outcome	Bloom's Taxonomy Level
CO 1	**Recall** the basic concepts of anatomy, physiology, cells, tissues, and body systems	**BT 1**
CO 2	**Explain** the structure and functions of skeletal, muscular, and joint systems	**BT 2**
CO 3	**Apply** knowledge of body systems such as respiratory, circulatory, and digestive systems in physical activity contexts	**BT 3**

CO 4	**Analyze** the functions of excretory, endocrine, nervous, and sensory systems in relation to physical performance	**BT 4**

Modules	Course Content	Periods
I	**Fundamentals of Anatomy & Physiology** 1.1 Meaning and Concept of Anatomy and Physiology and their application in the field of Physical Education and Sports. 1.2 Definition of Cells, Tissues, Organs and Organ Systems. 1.3 Microscopic structure, composition and functions of the cell. 1.4 Classification, structure and functions of tissues, organs and organ systems.	**15**
II	**Skeletal and Muscular System** 2.1 Skeletal System – Types of bones, their structure and functions. 2.2 Joint – Types and Classification of joints. 2.3 Muscular System – Meaning, Types, Structure, and functions of the Muscular system. 2.4 Overview of major muscles in the human body**.**	**15**
III	**Functional Body Systems** 3.1 Respiratory System – Organs of the respiratory system and types of respiration, Mechanism of Respiration and Functions of the Respiratory System. 3.2 Circulatory System – Structure & functions of the heart, Blood Vessels – Arteries, Veins and Capillaries, mechanism of blood circulation. 3.3 Digestive System – Organs, Mechanism and Functions of the digestive system.	**15**
IV	**Regulatory and Control Systems** 4.1 Excretory System – Organs of the excretory system and their structure and functions. 4.2 Endocrine System – Meaning of Glands, Endocrine glands, their location and functions. 4.3 Nervous System – Structure of Brain and spinal cord and functions of the nervous system. 4.4 Sensory system – kinaesthetic sense, visual and auditory sense, Temperature Regulation	**15**
	TOTAL	**60**

References :

1. Bourne, G.H. (1973). *"The Structure and Functions of Muscles".* Academic Press, London.

2. Chaurasia, B.D. (1979). *"Human Anatomy, Regional and Applied".* CBS Publishers and Distributors.

3. Gupta, A.P. (2010). *"Anatomy and Physiology".* Sumit Prakashan, Agra.

4. Gupta, M. & Gupta, M.C. (1980). *"Body and Anatomical Science".* Swaran Printing Press. New Delhi.

5. Jain, P., *"Basic Anatomy and Physiology of Exercise".* Khel Sahitya Kendra. New Delhi.

6. Lamb, L.S. (1982). *"Essentials of Exercise Physiology".* Surjeet Publications. New Delhi.

7. Pearce, E.C. (1962). *"Anatomy and Physiology for Nurses".* Faber & Faber Ltd. Publications, Saunders Company.

8. Morehouse, L.E. & Miller, J. (1976). *"Physiology of Exercise".* 7th Edition. St. Louis: The C.V. Mosby Company.

9. Moorthy, A.M. (2014). *"Anatomy, Physiology and Health Education".* Karaikudhi: Madalayam.

10. Guyton, A.C. (1996). *"Textbook on Medical Physiology".* 9th Edition, Philadelphia, U.S.A.

VAC : HEALTH AND WELLNESS	
SUBJECT CODE: BPE29V2101	**L-T-P = 2-1-0**
CREDIT UNITS: 03	**SCHEME OF EVALUATION: THEORY**

CREDIT DISTRIBUTION		
LECTURE/TUTORIAL	**PRACTICALS**	**EXPERIENTIAL LEARNING**
60	00	30

Course Objective: - The course aims to develop awareness and understanding of health, wellness, and lifestyle practices essential for overall well-being. It focuses on the physical, mental, and social dimensions of health, promoting healthy behaviours, nutritional knowledge, stress management, and positive lifestyle choices to enhance the quality of life among students.

CO	**Course Outcome**	**Bloom's Taxonomy Level**
CO 1	**Recall** the concepts of health, wellness, and components of health	**BT 1**
CO 2	**Explain** the factors affecting health, nutrition, and healthy lifestyle practices	**BT 2**
CO 3	**Apply** knowledge of health behaviour, stress management, and positive lifestyle choices in daily life	**BT 3**
CO 4	**Analyze** mental health, emotional well-being, and the role of social support in maintaining wellness	**BT 4**

Modules	Course Content	Periods
I	**Introduction to Health & Wellness** 1.1 Meaning and definition of health 1.2 Concept and components of wellness 1.3 Importance of health in everyday life 1.4 Factors affecting health and wellness 1.5 Components of health: physical, mental, social, spiritual and its relevance 1.6 Determinants of health (lifestyle, environment, heredity)	**10**
II	**Health Behaviour & Lifestyle** 2.1 Concept of healthy lifestyle & Personal hygiene 2.2 Fundamentals of Nutrition- Balanced diet, Macronutrients and Micronutrients (Protein, Carbohydrate, Fat, Vitamins, Mineral and water). 2.3 Substance abuse (tobacco, alcohol, drugs) 2.4 Promotion of positive lifestyle choices.	**10**
III	**Mental Health & Well-being** 3.1 Meaning of mental health 3.2 Stress: causes, symptoms, management 3.3 Emotional well-being- Implications of mind-body connections 3.4 Mental health vs mental illness (basic awareness) 3.5 Self-esteem and self-concept 3.6 Mindfulness and meditation 3.7 Digital well-being 3.8 Interpersonal relationships and social support 3.9 Mental well-being among students	**10**
	Total	**30**

Textbooks, Reference Books and Others

1. Insel, P. M., & Roth, W. T. Connect Core Concepts in Health.
2. Hoeger, W. W. K., Hoeger, S. A., Hoeger, C. I., & Fawson, A. L. Principles and Labs for Physical Fitness and Wellness.
3. Robbins, G., Powers, D., & Burgess, S. A. Wellness Way of Life.
4. Online Resources- Resources/e-books and e-learning portals

CREDIT DISTRIBUTION		
LECTURE/TUTORIAL	**PRACTICALS**	**EXPERIENTIAL LEARNING**
45	00	15

MAJOR SUBJECT: SPORTS ACTIVITY

SUBJECT CODE: BPE292M111	**L-T-P = 0-0-8**
CREDIT UNITS: 04	**SCHEME OF EVALUATION: PRACTICAL**

Course Objective: - The course aims to develop students' understanding and practical skills in Volleyball, Football, and Cricket. It covers basic knowledge of rules, techniques, and gameplay while enabling students to demonstrate skills and apply training principles in real situations. Students will also learn to analyze performance, improve techniques, and understand tactics, officiating, and teamwork, promoting overall sports competence and discipline.

CO	Course Outcome	Bloom's Taxonomy Level
CO 1	**Recall** rules, history, terminology, and basic concepts of Volleyball, Football, and Cricket	**BT 1**
CO 2	**Explain** skills, techniques, rules, and principles of gameplay and training	**BT 2**
CO 3	**Demonstrate** skills, apply techniques, and use strategies in practical/game situations	**BT 3**
CO 4	**Analyze** performance, identify errors, compare techniques, and evaluate strategies	**BT 4**

MODULE I: VOLLEYBALL

1. Foundation & Rules

- Historical development of Volleyball at National and International levels
- Rules and regulations of the game and latest amendments
- Impact of recent rule changes on the game
- Distinguished personalities in Volleyball (India & International) and awards
- Qualification system of major National and International tournaments

2. Skills of Volleyball

2.1 Fundamental Skills

- **Player Stance (Receiving & Passing):** Overhead Pass (Volley), Underarm Pass (Dig)
- **Service:** Underarm Service, Side Arm Service, Tennis Service, Round Arm Service

- **Lead-up Games:** Three Volleys, Three Digs
- **Spiking:** Straight Arm Spike, Round Arm Spike
- **Blocking:** Single Block

2.2 Advanced Skills

- **Passing:** Back Pass, Back Roll Volley, Back Roll Dig, Jump Pass, Side Roll Dig
- **Service:** Side Arm Floater**,** Overhead Floater
- **Spiking:** Cross Court Spike, Down the Line Spike
- **Blocking:** Double Block, Triple Block
- **Drive:** One-handed and Double-handed Drive (with dig combination)

3. Techniques and Application

- Classification and analysis of skills/techniques
- Identification of faults, causes, and corrections
- Exercises for development of volleyball skills
- Teaching stages of techniques
- Evaluation of techniques
- Recreational and lead-up activities
- Warm-up and cool-down exercises
- Training methods for development of motor abilities: Strength, Speed, Endurance, Flexibility, Coordination

4. Tactics and Application

- Philosophy of coaching in volleyball
- Preparation of coaching lessons
- Offensive strategies
- Defensive strategies
- Team building (on-court and off-court)

5. Officiating and Organization

- Volleyball court construction and markings

- Equipment and specifications
- Qualities and qualifications of officials
- Officiating mechanics: signals, whistle, score-sheet
- Organization of competitions (college, district, national level)

6. Talent Identification & Development

- Talent identification and development
- Planning and periodisation
- Preparation of training schedules
- Short-term and long-term training plans
- Periodisation in volleyball training
- General and specific fitness tests
- Skill and performance testing

7. Record Work & Practical

- Preparation of practical record/book including: History of the game, Skill analysis, Major national and international events
- Viva-voce
- Skill test
- Coaching lessons in Volleyball

References:

1) Anthony, C. and Lawrence, V. (2009). "Volleyball Player: Handbook. Friends Publication.

2) Ranganathan, P.P. (2003). "Volleyball: A Guide to Playing and Coaching". Friends Publication, New Delhi.

3) Sagar, S.K. (2001). "Play Better Volleyball". Sports Publication.

4) American Sport Education Program. (2007). "Coaching Youth Volleyball". Human Kinetics.

5) Wise, M. (1999). "Volleyball Drills for Champions". Human Kinetics.

6) Dhanraj, V.H. "Volleyball for Men and Women". Y.M.C.A. Publishing House.

7) Hawkins, V. (2018). "Rule Book of Games: Volleyball". Alpha Editions.

8) Murtaza, A. “Volleyball Skill Book”. Khel Sahitya Kendra.

MODULE II: FOOTBALL

1. Foundation & Rules

- Historical development of Football at National and International levels
- Rules and regulations of the game and latest amendments
- Impact of recent rule changes on the game
- Distinguished personalities in Football (India & International) and awards
- Qualification system of major National and International tournaments

2. Skills of Football

2.1 Fundamental Skills

- Kicking: Inside of the foot, Instep of the foot, Outer instep of the foot, Lofted kick
- Trapping: Rolling ball (inside, sole, instep), Bouncing ball (sole)
- Dribbling: Instep, Inside of the foot, Outer instep
- Heading: From standing, From running, From jumping
- Throw-in
- Feinting: Using lower limb, Using upper body
- Tackling: Simple tackling, Slide tackling
- Goalkeeping: Collection of balls, Ball clearance (kicking, throwing, deflecting)

2.2 Advanced Skills

- Kicking: Chip, In-swing and out-swing, Volley (low drive, back volley, scissor volley), Half volley
- Ball Reception & Control: Rolling ball with inside/outside and direction change, Trapping bouncing ball with abdomen and lower leg, Receiving rolling ball with direction change, Receiving flying ball (inside, instep, thigh, chest, head)
- Dribbling: Controlled dribbling, Fast dribbling, Straight dribbling, Zig-zag dribbling, Dribbling around/between obstacles
- Heading: From running and jumping, Long-distance heading, Downward heading

3. Techniques and Application

- Classification and analysis of skills/techniques
- Identification of faults, causes, and corrections
- Exercises for development of football skills
- Teaching stages of techniques
- Evaluation of techniques
- Recreational and lead-up activities
- Warm-up and cool-down exercises
- Training methods for development of motor abilities: Strength, Speed, Endurance, Flexibility, Coordination

4. Tactics and Application

- Philosophy of coaching in football
- Offensive strategies
- Defensive strategies
- Team building (on-field and off-field)

5. Officiating and Organization

- Football field construction and markings
- Equipment and specifications
- Qualities and qualifications of officials
- Officiating mechanics: whistle, signals, score-sheet
- Organization of competitions (college, district, national level)

6. Talent Identification & Development

- Talent identification and development
- Planning and periodisation
- Preparation of training schedules
- Short-term and long-term training plans
- Periodisation in football training
- General and specific fitness tests

- Skill and performance testing

7. Record Work & Practical

- Preparation of practical record/book including: History of the game, Skill analysis, Major national and international events
- Viva-voce
- Skill test
- Coaching lessons in football

Reference :

1. Wade, A. (1967). "The FA Guide to Training and Coaching". London: The Football Association.

2. Singh, G. "Football Quiz". Service Publishing House. New Delhi.

3. Saunders, T. "Play Better Soccer". London. Collins Glasgow.

4. Sharma, N.P. (2008). Skills and Techniques in Football". Khel Sahitya Kendra.

5. Bobby, M. (1985). "The Basic Soccer Guide". Collier Books.

6. Reilly, T. (2003). "Motion Analysis and Physiological Demands in Football". 1(2). 59-72.

MODULE III: CRICKET

1. Foundation & Rules

- Historical development of Cricket at National and International levels
- Rules and regulations of the game and latest amendments
- Impact of recent rule changes on the game
- Distinguished personalities in Cricket (India & International) and awards
- Qualification system of major National and International tournaments

2. Skills of Cricket

2.1 Fundamental Skills

- Batting: Grip, Stance, Backlift, Footwork (placement of feet), Contact with the ball
- Defensive Shots: Front foot defence: Back foot defence
- Bowling: Grip, Smooth run-up, Delivery stride, Follow-through
- Fielding: Stopping the ball, Throwing, High catching, Low catching
- Wicket-Keeping: Stance, Footwork, Hand positioning, Ball gathering

2.2 Advanced Skills

- Batting Strokes:Leg cut, Square cut, Cover drive, Straight drive, On drive, Hook shot, Pull shot, Sweep shot, Leg glance
- Bowling Skills: Outswing, Inswing, Off-cutter, Leg-cutter, Off-break, Leg-break, Googly, Bouncer, Beamer, Yorker

3. Techniques and Application

- Classification and analysis of skills/techniques
- Identification of faults, causes, and corrections
- Exercises for development of cricket skills
- Teaching stages of techniques
- Evaluation of techniques
- Recreational and lead-up activities
- Warm-up and cool-down exercises
- Training methods for development of motor abilities: Strength, Speed, Endurance, Flexibility, Coordination

4. Tactics and Application

- Philosophy of coaching in cricket
- Preparation of coaching lessons
- Offensive and defensive strategies (batting, bowling, field placement)
- Team building (on-field and off-field)

5. Officiating and Organization

- Cricket field construction and markings
- Equipment and specifications
- Qualities and qualifications of officials
- Officiating mechanics: signals, score-sheet

- Organization of competitions (college, district, national level)

6. Talent Identification & Development

- Talent identification and development
- Planning and periodisation
- Preparation of training schedules
- Short-term and long-term training plans
- Periodisation in cricket training
- General and specific fitness tests
- Skill and performance testing

7. **7. Record Work & Practical**

- Preparation of practical record/book including: History of the game, Skill analysis, Major national and international events
- Viva-voce
- Skill test
- Coaching lessons in Cricket

Reference:

1. Aneja, O.P., (2012). "How to Play Cricket". Prerna Prakashan. New Delhi.

2. Arora and Monika. (2005). "Cricket Coaching Manual". Sports Publications.

3. Bharadwaj and Arun (2008). "Coaching: Batting skills". Royal Colour Cartons.

4. Kutty, S. (2003). "Fielding Drills in Cricket". Sports Publication.

5. Srivastava, A.K. (2006). "How to Coach Bowling". Sports Publication.

6. Srivastava, V.K. (2007). Analysis of Cricket Skills". Sports Publication.

CREDIT DISTRIBUTION		
LECTURE/TUTORIAL	**PRACTICALS**	**EXPERIENTIAL LEARNING**

00	120	60

SEC 1: Formal Activities, Callisthenics, March Past

SUBJECT CODE: BPE292S111 **L-T-P = 0-0-6**

CREDIT UNITS: 03 **SCHEME OF EVALUATION: PRACTICAL**

Course Objective: - The course aims to develop students' coordination, rhythm, discipline, and physical fitness through formal activities, calisthenics, and march past. It enables students to understand and perform structured movements, apply rhythmic exercises, and demonstrate command-based drills effectively. Students will also develop the ability to analyze body movements, improve posture, and enhance synchronization and teamwork.

CO	Course Outcome	Bloom's Taxonomy Level
	Recall commands, exercise patterns, and names of activities	**BT 1**

CO 1		
CO 2	**Explain** sequences, rhythms, and movement techniques	**BT 2**
CO 3	**Perform** formal activities, calisthenics, and march past drills	**BT 3**
CO 4	**Analyze** posture, coordination, and synchronization of movements	**BT 4**

MODULE I: Formal Activities (with Light apparatus)

- Lathi exercises
- Lezium exercises
- Dumbbell exercises
- Hoops exercises
- Umbrella activities/exercises
- Ribbon activities/exercises
- Wand exercises

(Focus on rhythmic movements, coordination, and synchronized group performance using Light apparatus.)

MODULE II: Calisthenics

- **Exercises with verbal, whistle, drum, or musical beats**
    - Two counts
    - Four counts
    - Eight counts
    - Sixteen counts
- **Types of exercises**
    - Standing exercises
    - Sitting exercises
    - Jumping exercises
    - Moving exercises
- **Combination exercises**
    - Integration of standing, jumping, and moving exercises

MODULE III: Marching

- Standard English/Hindi Commands:
    - Attention

- Stand at Ease
- Stand Easy
- Right Turn
- Left Turn
- About Turn
- Right Incline
- Left Incline
- Dismiss
- Open Order March
- Right/Left Dress
- By the Left / By the Right
- Centre / Quick March
- Flag hoisting & flag unfurling

CREDIT DISTRIBUTION		
LECTURE/TUTORIAL	**PRACTICALS**	**EXPERIENTIAL LEARNING**
00	120	60

DETAILED SYLLABUS OF 2ND SEMESTER

MAJOR SUBJECT: EXERCISE PHYSIOLOGY

SUBJECT CODE: BPE292M201 **L-T-P = 3-1-0**

CREDIT UNITS: 04 **SCHEME OF EVALUATION: THEORY**

Course Objective: - The course aims to provide students with an understanding of the physiological principles underlying human movement and exercise performance. It focuses on muscle structure, energy metabolism, cardiovascular and respiratory responses, and the influence of environmental conditions on sports performance. The course enables students to apply exercise physiology concepts in training and performance enhancement.

CO	Course Outcome	Bloom's Taxonomy Level
CO 1	**Recall** the basic concepts of physiology, exercise physiology, muscle structure, and types of muscle fibres	**BT 1**
CO 2	**Explain** the mechanisms of muscle contraction, energy systems, and physiological responses during exercise	**BT 2**
CO 3	**Apply** knowledge of cardiovascular, respiratory, and training responses to understand exercise adaptation and sports performance	**BT 3**
CO 4	**Analyze** the effects of climatic conditions, altitude, and exercise training on physiological responses and performance	**BT 4**

Modules	Course Content	Periods
I	**<u>Introduction to Exercise Physiology and Structure of Muscle</u>** 1.1 Meaning and Definition of Physiology, Sport and Exercise Physiology. 1.2 Different types of muscles (Voluntary, Involuntary and cardiac). 1.3 Macro and Microstructure of Skeletal Muscle, 1.4 Muscle Fibre Types (Red, White, and Intermediate).	**15**
II	**<u>Bioenergetics and Muscle Contraction</u>** 2.1 Types of muscle contraction 2.2 Fuel for Muscular Work (ATP). 2.3 Aerobic and anaerobic metabolism 2.4 Aerobic and Anaerobic systems during rest and exercise. 2.5 Theories of muscular contraction and the sliding filament theory.	**15**
III	**<u>Conditioning and Training on circulatory and respiratory Systems</u>** 3.1 Cardiac Cycle, Stroke Volume, Cardiac Output, factors affecting heart rate and Cardiac Hypertrophy. 3.2 Effect of Exercises and Training on the Cardiovascular System. 3.3 Mechanism of Breathing, Respiratory muscles, Minute ventilation, Diffusion of gases, Oxygen Debt, Lung Volumes and Capacities, Second Wind. 3.4 Effects of exercises and training on the respiratory system.	**15**
IV	**<u>Climatic Conditions and Sports Performance</u>** 4.1 Effects of Variation in Temperature and Humidity on Sports Performance 4.2 Thermoregulation, Sports performance in hot climates and cool climates. 4.3 Meaning and definition of altitude training, types of altitude training and Altitude sickness in sports. 4.4 Effects of altitude training on sports performance.	**15**
	TOTAL	**60**

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REFERENCE

1. Bob Murray, *Practical Guide to Exercise Physiology*, Human Kinetics,2016

2. Jonathan K. Ehrman, *Advanced Exercise Physiology: Essential Concepts and Applications*, Human Kinetics,2017

3. Jack H. Wilmore, *Physiology of Sports and Exercise,* Human Kinetics, 1994.

4. Tiwari Sandhya, *Exercise Physiology,* Sports Publication, New Delhi, 2021

5. Smith Dinesh L., *Advanced Cardiovascular Exercise Physiology,* Human Kinetics, 2010

6. W. Larry Kenney, *Physiology of Sport and Exercise*, Sixth Edition, Human Kinetics, 2015

7. Yadav Hiralal, *Physiological Responses and Recovery Pattern in Athletics*, Sports Publication, New Delhi,2020

Online Resources- Resources/e-books and e-learning portals

CREDIT DISTRIBUTION		
LECTURE/TUTORIAL	**PRACTICALS**	**EXPERIENTIAL LEARNING**
60	00	30

MAJOR SUBJECT: HEALTH EDUCATION & ENVIRONMENTAL STUDIES

SUBJECT CODE: BPE292M202 **L-T-P = 3-1-0**

CREDIT UNITS: 04 **SCHEME OF EVALUATION: THEORY**

Course Objective: - The course aims to provide students with knowledge and understanding of health education, health problems, and environmental issues affecting individuals and society. It focuses on developing awareness about health promotion, personal hygiene, school health services, environmental conservation, natural resources, and sustainable development practices**.**

CO	**Course Outcome**	**Bloom's Taxonomy Level**

CO 1	**Recall** the basic concepts, dimensions, principles, and determinants of health and environmental studies	**BT 1**
CO 2	**Explain** health problems, health services, environmental concepts, and conservation practices	**BT 2**
CO 3	**Apply** knowledge of health education, hygiene practices, and environmental awareness in daily life and educational settings	**BT 3**
CO 4	**Analyze** environmental issues, resource management strategies, and the role of health education in promoting sustainable development	**BT 4**

Modules	Course Content	Periods
I	**HEALTH EDUCATION** 1.1 Concept, Dimensions, Spectrum and Determinants of Health 1.2 Definition of Health, Health Education, Health Instruction, Health Supervision 1.3 Aim, objectives and Principles of Health Education 1.4 Health Service and guidance instruction in personal hygiene	**15**
II	**Health Problems in India** **2**.1 Communicable and Non-Communicable Diseases 2.2 Obesity, Malnutrition, Adulteration in food, Environmental sanitation, Explosive Population, 2.3 Personal and Environmental Hygiene for schools 2.4 Objective of school health service, Role of health education in schools 2.5 Health Services: Care of skin, Nails, Eye, Nutritional service, Health appraisal, Health record, Healthful school environment, first- aid and emergency care, etc.	**15**
III	**Environmental Science**: 3.1 Definition, Scope, need for and Importance of environmental studies 3.2 Concept of environmental education, Historical background of environmental education, 3.3 Celebration of various days in relation to the environment. 3.4 Plastic recycling & prohibition of plastic bag/ cover. 3.5 Role of school in environmental conservation and sustainable development	**15**
IV	**Natural Resources and related environmental issues:** 4.1 Water resources, food resources and Land resources 4.2 Definition, effects and control measures of: 4.3 Air Pollution, Water Pollution, Soil Pollution, Noise Pollution, Thermal Pollution 4.4 Management of environment and Govt. policies, Role of pollution control board.	**15**
	Total	**60**

References:

- Agrawal, K.C. (2001). Environmental biology.Bikaner: Nidhi Publishers Ltd.
- Frank, H. &Walter, H. (1976). Turner's school health education. Saint Louis: The C.V. Mosby Company.
- Nemir, A. (n.d.). The school health education. New York: Harper and Brothers.
- Odum, E.P. (1971). Fundamentals of ecology. U.S.A.: W.B. Saunders Co.

CREDIT DISTRIBUTION		
LECTURE/TUTORIAL	PRACTICALS	EXPERIENTIAL LEARNING
60	00	30

VAC : HOLISTIC YOGA EDUCATION

SUBJECT CODE: BPE292V201 **L-T-P = 2-0-0**

CREDIT UNITS: 02 **SCHEME OF EVALUATION: THEORY**

Course Objective: - The course aims to provide students with a comprehensive understanding of the philosophy, principles, and practices of Yoga. It focuses on the concepts of Ashtanga Yoga, yogic practices, meditation, and their applications for physical, mental, and physiological well-being. The course enables students to understand the role of Yoga in health promotion, stress management, and sports performance enhancement.

CO	Course Outcome	Bloom's Taxonomy Level
CO 1	**Recall** the meaning, history, concepts, and fundamental principles of Yoga	**BT 1**
CO 2	**Explain** various yogic practices including Shatkarma, Asanas, Pranayama, Mudras, Bandhas, and their benefits	**BT 2**
CO 3	**Apply** yogic techniques and meditation practices for improving physical and mental well-being	**BT 3**
CO 4	**Analyze** the role of Yoga in managing health problems, physiological functions, and sports performance	**BT 4**

Modules	Course Content	Periods
I	**<u>INTRODUCTION TO YOGA & MEDITATION</u>** 1.1 Meaning, definitions and historical background of yoga. 1.2 Aims, objectives, importance and scope of yoga in the modern era. 1.3 Meditation: Meaning, definition, types, meditation environment,	**10**

	techniques and benefits of meditation. 1.4 Applications and misconceptions of yoga in modern society. 1.5 Introduction to Patanjali Yoga Sutra. 1.6 Concept of Ashtanga Yoga	
II	**<u>SHATKARMA, ASANAS, PRANAYAM, MUDRAS AND BANDHAS</u>** 2.1 Shatkarmas: Meaning, definition, types, techniques, precautions and benefits. 2.2 Asanas: Meaning, definition, types, techniques, precautions and benefits. 2.3 Pranayama: Meaning, definition, types, techniques, precautions and benefits. 2.4 Mudras: Meaning, definition, types, techniques, precautions and benefits. 2.5 Bandhas: Meaning, definition, types, techniques, precautions and benefits. 2.6 Suryanamaskar: Meaning, definition, Suryanamaskar mantra and its meaning, technique. Precautions and benefits.	**10**
III	**<u>MEDITATION AND YOGA FOR COMMON HEALTH PROBLEMS</u>** 3.1 Role of yoga in psychological preparation of athletes. 3.2 Yogic practices for the common health problems: Constipation, diabetes, obesity, cervical arthritis, backache, sciatica pain, indigestion, snoring, eye disorders and migraine. 3.3 Effect of yoga on the different physiological systems: Glandular, circulatory, skeletal, digestive, nervous, respiratory, excretory and reproductive systems.	**10**
	Total	**30**

References:

1. Anatharaman, T.N. (1996*). "Ancient Yoga and Modern Science Project of History of Indian Sciences, Philosophy and Culture".*

2. Arya, K. (2011). *"Yogic Science".* Friends Publication.

3. Arya, K. (2013). "*Yogic Education".* Friends Publication.

4. Debnath, K. K. (2010). *"Yogic Sciences".* Friends Publication.

5. Rajjan, S.M. (1985). *"Yoga Strengthening and Relaxation for Sportsmen*". Allied Publishers.New Delhi.

6. Brow, F.Y. (2000). *"How To Use Yoga".* Sports Publication. New Delhi.

CREDIT DISTRIBUTION		
LECTURE/TUTORIAL	**PRACTICALS**	**EXPERIENTIAL LEARNING**
30	00	15

MAJOR SUBJECT: SPORTS ACTIVITY	
SUBJECT CODE: BPE292M211	**L-T-P = 0-0-8**
CREDIT UNITS: 04	**SCHEME OF EVALUATION: PRACTICAL**

Course Objective: - The course aims to develop students' practical knowledge and proficiency in Athletics (Track Events), Table Tennis, and Pickleball. It focuses on understanding rules, techniques, skills, tactics, officiating, and training methods of different sports. The course enables students to apply sport-specific skills, analyze performance, develop coaching abilities, and understand talent identification and development processes.

CO	Course Outcome	Bloom's Taxonomy Level
CO 1	**Recall** the history, rules, regulations, and fundamental concepts of Athletics, Table Tennis, and Pickleball	**BT 1**
CO 2	**Explain** the techniques, skills, training methods, and performance requirements of different sports	**BT 2**
CO 3	**Apply** sport-specific skills, tactics, fitness training methods, and officiating procedures in practical situations	**BT 3**
CO 4	**Analyze** sports performance, identify technical errors, apply corrective measures, and evaluate training and talent development strategies	**BT 4**

MODULE I: ATHLETICS (TRACK EVENTS)

(Sprints, Middle Distance, Long Distance, Relays, Hurdles and Race Walk)

1. Foundation & Rules

- Introduction and historical development of Track Events at National and International levels
- Important National and International tournaments in Track Events
- Rules and regulations of Track Events and latest amendments
- Impact of recent rule changes on the events
- World records in Track Events
- Notable personalities in Track & Field and National and International awards

2. Fundamental Skills and Techniques

2.1 Starting Techniques: Standing Start, Crouch Start, Flying Start and its variations, Proper use of starting blocks and its advantages

2.2 Finishing Techniques: Run Through, Forward Lunge, Shoulder Shrug

2.3 Relay Skills

- Fundamental skills of relay races: 4 × 100 m Relay, 4 × 400 m Relay
- Baton exchange techniques, Understanding of relay exchange zones

2.4 Hurdle Skills

- Fundamental skills and techniques of hurdle races
- Measurement and arrangement of hurdles for men and women
- Various age categories of hurdles for men & women

2.5 Race Walk: Fundamental skills and techniques of race walking

3. Training and Application

- Warm-up and cool-down exercises for various Track Events
- Drills and conditioning exercises for Track Events
- Training methods for development of motor abilities: Strength, Speed, Endurance, Flexibility, Coordination

4. Officiating and Organization

- Construction and marking of standard tracks: 400 m Track, 200 m Track
- Technical rules and duties of Track Officials
- Mechanism of officiating: Score sheets, Signals, Whistle
- Officiating procedures in Track Events
- Organization of Track competitions

5. Record Work and Practical

- Preparation of practical record/notebook including: History of Track Events, Skill analysis of events, Important National and International events
- Viva-Voce
- Skill Test

References:

1. Schneider, R.C., (2009). *"Ethics of Sports and Athletes*": Theory, Issues and Application.Wolters Kluwer Health.

2. Zeigler, E.F., & Speath, M.J. (1975). "*Administrative Theory and Practice in Physical Education and Athletics".*

3. Bompa, T.O. and Buzzicheli, C., (2018). *"Periodisation: Theory and Methodology of Training".* Human Kinetics.

4. Bompa, T. and Bompa, T.O., (2005). *"Periodization Training for Sports*". 2nd Edition, Human Kinetics.

5. Singh, H. (1984). *"Sports Training: General Theory and Methods*". Netaji Subhash National Institute of Sports.

MODULE II: TABLE TENNIS

1. Foundation & Rules

- Historical development of Table Tennis at National and International levels
- Rules and regulations of the game and latest amendments
- Impact of recent changes in rules and their influence on the game
- Distinguished personalities related to Table Tennis in India and abroad and their awards
- Qualification system of major National and International tournaments

2. Skills of Table Tennis

2.1 Fundamental Skills

- **Grip:** Shake-hand/Tennis grip, Forehand grip, Backhand grip, Penholder grip
- **Service: Forehand service:** Forward spin, Backward spin **Backhand service:** Forward spin, Backward spin
- **Spin services:** Side spin, Top spin, Back spin
- **Strokes:** Push, Chop, Drive (Top spin), Half volley, Smash, Drop shot, Balloon shot, Flick shot, Loop drive

2.2 Footwork and Position

- Stance and ready position
- Basic footwork techniques
- Movement and positioning on the court

3. Techniques and Application

- Classification and analysis of skills/techniques
- Identification of faults in skills, their causes, and corrective measures
- Exercises for development of fundamental Table Tennis skills
- Teaching progression and stages of techniques
- Evaluation of techniques

- Recreational and lead-up activities in Table Tennis
- Warm-up and cool-down exercises for Table Tennis
- Training methods for development of motor abilities: Strength, Speed, Endurance, Flexibility, Coordination

4. Tactics and Application

- Development of offensive and defensive strategies
- Match strategies and game planning
- Singles and doubles tactics
- Team building: On-court, Off-court

5. Officiating and Organization

- Dimensions and marking of Table Tennis table
- Equipment and specifications
- Qualities and qualifications of Table Tennis officials
- Mechanism of officiating: Signals, Score sheets, Cards, Scoreboard
- Organization of competitions: College level, District level, National level

6. Talent Identification & Development

- Detection and development of talent in Table Tennis
- Planning and periodisation of training
- Preparation of training schedules for players
- Short-term and long-term training plans
- Periodisation in player training
- General and specific fitness tests
- Skill and performance assessment tests

7. Record Work & Practical

- Preparation of practical record/notebook including: History of Track Events, Skill analysis of events, Important National and International events
- Viva-Voce
- Skill Test

Reference :

1. *"Manual of Table Tennis*" - International Table Tennis Federation (2023).

2. Rai, G. (2016). *"The Art of Table Tennis".* ISBN: 13:9788178799483.

3. Finn, K., (2021). *"Peak Performance in Table Tennis".* Meyer & Meyer Publishing.

4. Verma, H. (2016). *"How to Play Table Tennis*". ISBN: 13:9789381867198.

5. Jain, D. (2001). *"Table Tennis: Teaching and Coaching.* Khel Sahitya Kendra. New Delhi.

6. Narang, P. (2005). *"Plan and Learn Table Tennis*". Khel Sahitya Kendra. New Delhi.

MODULE III: PICKLEBALL

1. Foundation & Rules

- Historical development of Pickleball at National and International levels
- Growth and development of Pickleball as a modern sport
- Rules and regulations of the game and latest amendments
- Court dimensions, equipment specifications, scoring system, and basic regulations
- Important National and International tournaments
- Distinguished personalities related to Pickleball and awards
- Qualification system of major National and International tournaments

2. Skills of Pickleball

2.1 Fundamental Skills

- Grip and Ready Position: Basic paddle grip, Forehand grip, Backhand grip, Ready position and court movement
- Service: Forehand service, Backhand service, Service placement and accuracy
- Return of Service: Forehand return, Backhand return, Controlled return shots.
- Ground Strokes: Forehand drive, Backhand drive, Basic rally strokes.
- Volley Skills: Forehand volley, Backhand volley, Block volley
- Net Play: Dink shot, soft shots near the non-volley zone (Kitchen area**)**

2.2 Advanced Skills

- Advanced Shot: Third shot drop, Lob shot, Smash, Cross-court shots, Passing shots
- Spin Techniques: Top spin, Back spin, Side spin
- Movement Skills: Footwork patterns, Lateral movement, Court coverage and recovery position

3. Techniques and Application

- Classification and analysis of skills/techniques
- Identification of faults, causes, and corrective measures
- Exercises for development of fundamental Pickleball skills
- Teaching stages of techniques
- Evaluation of skills and performance
- Recreational and lead-up activities in Pickleball
- Warm-up and cool-down exercises
- Training methods for development of motor abilities: Strength, Speed, Endurance, Flexibility, Coordination

4. Tactics and Application

- Basic principles of Pickleball strategy
- Offensive and defensive strategies
- Singles and doubles game strategies
- Court positioning and movement strategies
- Shot selection and rally construction
- Team coordination and communication
- Match planning and game analysis

5. Officiating and Organization

- Construction and marking of Pickleball court
- Equipment required and specifications
- Qualities and qualifications of Pickleball officials
- Mechanism of officiating: Signals, Score sheets, Cards, Scoreboard
- Organization of competitions: College level, District level, National level

6. Talent Identification & Development

- Detection and development of talent in Pickleball
- Physical and skill-related components for player development
- Planning and periodisation of training
- Preparation of training schedules for players
- Short-term and long-term training plans

- General and specific fitness tests
- Skill and performance assessment tests

7. Record Work & Practical

- Preparation of practical record/notebook including: History of Track Events, Skill analysis of events, Important National and International events
- Viva-Voce
- Skill Test

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LECTURE/TUTORIAL	PRACTICALS	EXPERIENTIAL LEARNING
00	120	60

SEC: YOGA TECHNIQUES AND TRAINING

SUBJECT CODE: BPE292S211 **L-T-P = 0-0-4**

CREDIT UNITS: 02 **SCHEME OF EVALUATION: PRACTICAL**

Learning Objective: The course aims to develop practical proficiency in yogic techniques and training methods. It enables students to perform various asanas, pranayama, mudras, bandhas, shatkarmas, and prayer practices with proper techniques and precautions. The course focuses on improving physical flexibility, mental concentration, breathing control, and overall well-being through systematic yoga practices.

CO	Course Outcome	Bloom's Taxonomy Level

CO 1	**Recall** the basic concepts, names, and techniques of yogic practices including prayers, asanas, mudras, bandhas, and shatkarmas	**BT 1**
CO 2	**Explain** the methods, benefits, and precautions of various yoga techniques	**BT 2**
CO 3	**Demonstrate** different asanas, pranayama, mudras, bandhas, and cleansing practices with proper technique	**BT 3**
CO 4	**Analyze** the effects of yogic practices on physical fitness, mental well-being, and overall health	**BT 4**

MODULE I: PRAYERS

- **Introduction and practice of yogic prayers:**
    - Paramatma Vandana
    - Gayatri Mantra
    - Guru Vandana

MODULE II: ASANAS

1. Types of Asanas: Practice of various asanas in different positions: Sitting Position, Standing Position, Supine Position, Prone Position

2. Surya Namaskar: Technique and sequence of Surya Namaskar, Benefits and precautions

3. Meditative Postures: Samasana, Swastikasana, Siddhasana, Ardha Padmasana, Padmasana, Vajrasana

4. Resting Postures: Shavasana, Makarasana

5. Cultural Postures

- **Standing Asanas:** Tadasana, Vrikshasana, Garudasana, Utkatasana, Padahastasana, Ardha Chakrasana, Ardha Kati Chakrasana, Trikonasana, Parsva Konasana, Veerabhadrasana, Parshva Uttanasana
- **Sitting Asanas:** Janu Sirsasana, Akarna Dhanurasana, Ardha Kurmasana, Mandukasana, Simhasana, Sasankasana, Vakrasana, Ardha Matsyendrasana
- **Supine Asana:** Uttanpadasana, Pawanmuktasana, Sarvangasana, Halasana, Matsyasana
- **Prone Asanas:** Ardha Shalabhasana, Dhanurasana, Naukasana, Chakrasana

MODULE III: MUDRA AND BANDHA

1. Bandhas : Jalandhara Bandha, Uddiyana Bandha, Mula Bandha

2. Mudras: Ashwini Mudra, Mahabandha, Mahabheda Mudra

MODULE IV: SHATKARMA AND PRANAYAMA

1. Shatkarmas

- Neti: Jala Neti Sutra/Catheter Neti
- Dhauti: Agnisar, Vamana Dhauti
- Trataka (Sthula)
- Kapalbhati (Vatkrama)
- Basti
- Nauli

2. Pranayama and Breathing Practices: Sectional Breathing, Yogic Breathing, Nadi Shuddhi, Anulom-Vilom, Bhramari

References :

1. Anatharaman, T.N. (1996). "Ancient Yoga and Modern Science Project of History of Indian Sciences, Philosophy and Culture".

2. Arya, K. (2011). "Yogic Science". Friends Publication.

3. Arya, K. (2013). "Yogic Education". Friends Publication.

4. Debnath, K. K. (2010). "Yogic Sciences". Friends Publication.

5. Horovitz, E. G., and Elgelid, S. (2015). "Yoga Therapy: Theory and Practice". Routledge

6. Kotecha, R. and Vaidya Rajesh. (2016). "A Beginner's Guide to Ayurveda". Chakrapani Publications.

7. Kumar and Dr Kamakhya (2008). "Super Science of Yoga". Standard Publications.

8. Leslie, K. and Matthews, A., (2011). "Yoga Anatomy". Human Kinetics.

9. Nathial, M. S. (2013). "Yogic Education". Friends Publication.

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